How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

Anonymous Authors[1]

Abstract

Task descriptions and demonstrations both convey information about a task, but their relative value is difficult to quantify. We define the predictive effective sample size (ESS) of a description as the number of additional demonstrations required to match its reduction in prediction error. In a Bayesian model of in-context linear regression, we distinguish a description's specificity—the precision of the task information it provides when relevant—from its reliability—the probability that this information applies. Reliable descriptions can replace many demonstrations, with a benefit that depends on task dimension as well as precision. Any fixed positive probability of irrelevance, however, bounds their ESS when no demonstrations are available. Transformers meta-trained on this task approach the Bayesian benchmark and learn how much to trust descriptions from their training distribution. We then test pretrained language models on numerical prediction tasks with explicit probabilistic descriptions and on a semantic classification task. Without reasoning, the models respond weakly to stated precision; with reasoning, several approach the Bayesian reference. With reasoning, they also update a description's reliability from the demonstrations when the prompt states what follows if the description is wrong, but not when it leaves this open. Both behaviors depend on how explicitly the prompt specifies the description's statistical meaning. ESS thus provides a common scale for the information a description offers and the benefit a learner obtains from it.

1. Introduction

An in-context learner can learn about a task in two ways: a prompt can describe what to do, or it can demonstrate the desired behavior. Most practical prompts combine the two (Brown et al., 2020), and task descriptions also appear in system instructions and the files used to configure coding agents (Chatlatanagulchai et al., 2025). Yet we lack a common scale for comparing the information supplied by a description with that supplied by examples. When should a precise description replace many demonstrations? When can examples help a learner recognize that a description does not apply?

Empirical findings suggest that the answers depend on both the description and the learner. Prompts can substitute for substantial labeled data in prompt-based fine-tuning (Le Scao & Rush, 2021), and written summaries can replace demonstrations (Honda et al., 2025). Other studies find weak sensitivity to instruction relevance (Webson & Pavlick, 2022), excessive reliance on demonstrations relative to explicit instructions (Gupta et al., 2025), or limited benefits from repository instruction files (Gloaguen et al., 2026). These settings differ, but they raise a shared measurement problem: how much task information does a description provide, and how much of it does the model use?

We address this problem within the Bayesian view of in-context learning (ICL), which treats the prompt as evidence about a latent task (Xie et al., 2022). A description induces a distribution over tasks before the demonstrations are observed. We characterize this distribution through two properties. *Specificity* measures how tightly a relevant description constrains the task. *Reliability* measures the probability that the description is relevant in the first place. A broad but correct description and a precise description of the wrong task create different inference problems: the latter requires the learner to assess whether the demonstrations support the description.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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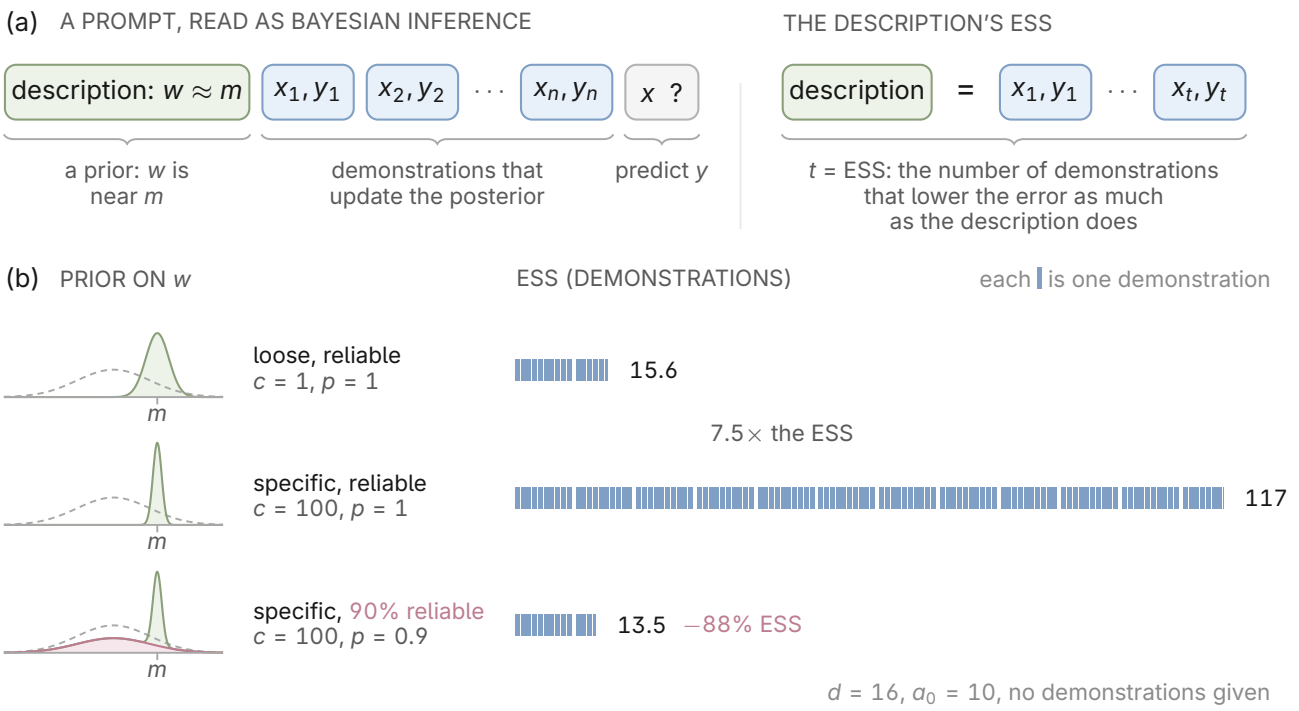


Figure 1. **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w; demonstrations supply additional evidence. Its effective sample size (ESS) is the number of additional demonstrations needed to achieve the same prediction error. (b) Baseline and description-conditioned task distributions, shown schematically, with their Bayesian ESS at $n = 0$, $d = 16$, and $a_0 = 10$. Specificity c is the precision of a relevant description; reliability p is its probability of relevance. Pink denotes an irrelevant description. Increasing specificity can greatly increase ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit without demonstrations.

To compare descriptions with demonstrations, we adapt the effective sample size of an informative prior (Morita et al., 2008; Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021). Our *predictive ESS* is the number of additional demonstrations that a learner without the description needs to match the learner's error with it. The definition applies at any number of demonstrations and to any predictor. It therefore distinguishes the benefit available to a Bayesian learner from the benefit realized by a trained model.

We develop this framework in Gaussian linear regression, where description information and prediction error can be analyzed directly. We then study two ways a learner can acquire that information. Meta-trained Transformers learn the specificity and reliability of description tokens from their training distribution. Pretrained language models instead receive explicit probabilistic descriptions in text. This second setting isolates whether models use stated uncertainty when combining a description with numerical demonstrations; it does not require assigning an exact probability distribution to an unrestricted natural-language instruction.

Our contributions are threefold:

- **A predictive measure of description value.** We define description ESS by matching prediction errors, using each learner's own learning curve as the reference (Definition 1).
- **An analysis of specificity and reliability.** We derive risk formulas and ESS limits for Gaussian task descriptions. Reliable descriptions benefit from both precision and coverage of task dimensions. Unreliability imposes a finite ceiling without demonstrations, while demonstrations can help assess relevance (Sections 3–4).
- **Tests of learned and stated task information.** Meta-trained Transformers approach the Bayesian benchmark. In pretrained language models, reasoning substantially improves the use of stated precision, and recovers Bayesian reliability updating when the prompt states what a wrong description implies, in both numerical and semantic tasks. Both depend on how explicitly the prompt specifies the description's statistical meaning (Sections 5–6).

2. Framework

Tasks and demonstrations. We use the standard noisy in-context linear regression setting (Garg et al., 2022; Raventós et al., 2023). A task is a vector $w \in \mathbb{R}^d$ drawn from the baseline prior $\mathcal{N}(0, a_0 I_d)$. A prompt contains n demonstrations

$D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x, where

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

The parameter $a_0 > 0$ controls variation across tasks. Numerical illustrations in Sections 3–4 use $d = 16$ and $a_0 = 10$.

Descriptions as uncertain task information. A description supplies a center $m \in \mathbb{R}^d$. Its *specificity* is a precision $c \geq 1/a_0$, and its *reliability* is a probability $p \in [0, 1]$. Write $r = (ca_0)^{-1} \in (0, 1]$ for the fraction of baseline variance remaining when the description is relevant. With an unobserved relevance indicator $Z \sim \text{Bernoulli}(p)$, independent of m, we generate

$$\begin{aligned} m &\sim \mathcal{N}(0, (1-r)a_0 I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, ra_0 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d). \end{aligned}$$

Thus the description induces the mixture prior

$$w \mid m \sim p\mathcal{N}(m, c^{-1} I_d) + (1-p)\mathcal{N}(0, a_0 I_d),$$

as in robust mixture priors for potentially incompatible information (Schmidli et al., 2014). Both relevance branches have the same marginal task distribution, $w \sim \mathcal{N}(0, a_0 I_d)$. Consequently, the comparison with and without a description does not change the task population.

A description is *reliable* when $p = 1$ and *unreliable* when $p < 1$. Greater c means greater specificity conditional on relevance. The model separates uncertainty within the described region from uncertainty about whether that region applies. It does not assume that every natural-language description can be summarized by these two scalars.

Prediction error. For a prediction $\hat{y}$ of the query response, we measure error against its conditional mean:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{1}$$

This excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, let $R^{\text{desc}}(n)$ and $R^{\text{demo}}(n)$ denote its errors with and without a description. We write $R^{\text{desc}}_{p=1}(n)$ for the reliable case.

Definition 1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ such that $R^{\text{demo}}(t) = R^{\text{desc}}(n)$. When this crossing exists, the ESS of the description is

$$\text{ESS}_n = n^* - n.$$

We interpolate R^{demo} between available demonstration counts: linearly in t for the theory and meta-trained models, and in $\log t$ for language models. For measured curves, we use the first crossing within the observed range and report out-of-range values as bounds. The language-model conventions are detailed in Appendix C. If $R^{\text{desc}}(n) > R^{\text{demo}}(t)$ for all $t \geq 0$, we report $\text{ESS}_n \leq -n$.

ESS measures a horizontal displacement of a learning curve: $\text{ESS}_n = 10$ means that n demonstrations plus the description match the error of $n+10$ demonstrations without it. For a decreasing baseline curve, a negative value indicates increased error at the same demonstration count. Each model is evaluated against its own curve, so ESS measures the description's value *to that learner*; a large ESS alone does not imply low absolute error. Under the correctly specified Bayesian model, descriptions cannot increase expected risk, and ESS is nonnegative.

3. The Value of a Reliable Description

For $p = 1$, the Bayesian predictor uses the posterior mean of $w^\top x$. Its risk is the expected trace of the posterior covariance of w. Let X_n contain the demonstration inputs as rows and write $W_n = X_n^\top X_n$. Then

$$\begin{aligned} R^{\text{demo}}(n) &= \mathbb{E}\,\text{tr}(a_0^{-1} I_d + W_n)^{-1}, \\ R^{\text{desc}}_{p=1}(n) &= \mathbb{E}\,\text{tr}(cI_d + W_n)^{-1}. \end{aligned}$$

The description replaces baseline precision a_0^{-1} with c. Its predictive value is not, however, determined by this precision difference alone.

Proposition 1 (Reliable descriptions)**.** *Let $p = 1$.*

(i) With $r \in (0, 1]$ fixed, $\text{ESS}_0 \geq (1-r)d$, and $\text{ESS}_0 \to (1-r)d$ as $a_0 \to \infty$.

(ii) With positive integer c fixed, $\text{ESS}_0 \to c + d + 1$ as $a_0 \to \infty$.

The two limits describe different regimes (proof in Appendix A.1). In the first, the description removes a fixed fraction of uncertainty in every direction. Each of the first d demonstrations constrains one direction, so reducing variance across all directions can save approximately $(1-r)d$ demonstrations. In the second, the description retains a fixed absolute precision while the baseline prior becomes diffuse. Without a description, the limiting risk is $d/(n-d-1)$ for $n > d+1$ (Muirhead, 1982); the description alone has risk d/c. These risks match at $n = c + d + 1$. The integer restriction makes the crossing coincide with a measured count under linear interpolation.

This $d + 1$ contribution reflects the cost of learning a d-dimensional task from random inputs. Matching average information would count only the added precision $c - a_0^{-1}$; matching predictive error also accounts for the random design. We refer to the additional contribution as the *dimension term*. It is substantial when c is small relative to d,

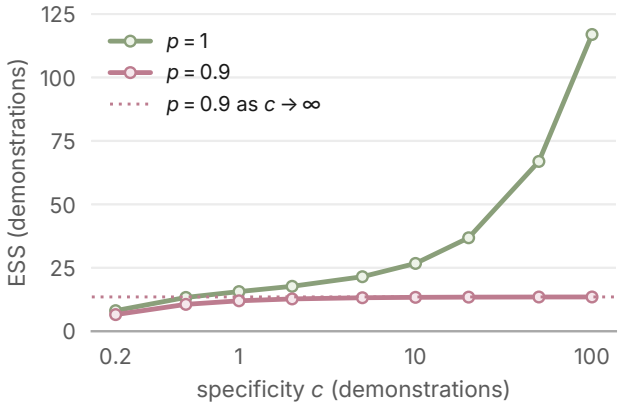


Figure 2. **Reliability limits the benefit of greater specificity without demonstrations.** Bayesian ESS_0 versus specificity c, for $p = 1$ and $p = 0.9$, at $d = 16$ and $a_0 = 10$. The dotted line is the finite ceiling for $p = 0.9$ as $c \to \infty$. Estimates use eight batches of 4,000 prompts; standard errors are below 0.02 demonstrations.

whereas precision dominates when c is large. This interpretation concerns the stated limits, rather than an exact additive formula for every finite a_0 and n.

At $d = 16$ and $a_0 = 10$, a reliable description with $c = 1$ has ESS 15.6, compared with the lower bound $(1 - r)d = 14.4$. Increasing c to 100 raises ESS to approximately 117 (Figure 2). As demonstrations accumulate, the dimension benefit disappears: $\mathrm{ESS}_n \to c - a_0^{-1}$ (Corollary 3). For example, after 100 demonstrations the $c = 1$ description is worth 1.1 demonstrations, close to its added precision of 0.9.

Coverage matters as well as precision. A description may constrain only $k < d$ directions. For any fixed precision $c > 0$, its zero-demonstration ESS approaches k as $a_0 \to \infty$; increasing that fixed precision does not remove uncertainty in the other directions. With many demonstrations, its ESS instead approaches $k(c - a_0^{-1})/d$ (Proposition 4). A precise partial description can therefore become more valuable as demonstrations resolve what it leaves unspecified. At $d = 16$, $k = 8$, and $c = 100$, ESS increases from 8.1 without demonstrations to 36 with 100 demonstrations (Table 1).

4. The Cost of Unreliability

When $p < 1$, the learner must also infer whether the description applies. Let μ_1 and μ_0 denote the posterior means conditional on relevance and irrelevance, and let $\pi_n = P(Z = 1 \mid m, D_n)$ be the posterior relevance probability. The Bayesian prediction is

$$\hat{y} = \left[\pi_n \mu_1 + (1 - \pi_n)\mu_0\right]^\top x.$$

Demonstrations affect both the component predictions and their relative weight.

Proposition 2 (Risk under uncertain relevance)**.** *For every $n \geq 0$,*

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1 - p)R^{\mathrm{demo}}(n) + \mathcal{V}_n, \tag{2}$$

where

$$\mathcal{V}_n = \mathbb{E}\Big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\Big] \geq 0.$$

With no demonstrations,

$$R^{\mathrm{desc}}(0) = \left[1 - p^2(1 - r)\right]a_0 d. \tag{3}$$

Equation (2) separates the error of an oracle that knows Z from the cost of uncertainty about relevance. The oracle uses the description when it applies and otherwise uses the baseline prior. The remaining term, $\mathcal{V}_n$, is the disagreement between these two predictions, weighted by uncertainty about which is appropriate. Without demonstrations, the oracle removes a fraction $p(1 - r)$ of baseline risk; a learner that must infer relevance removes only $p^2(1 - r)$. The proof follows from the law of total variance (Appendix A.2).

Corollary 3 (ESS under uncertain relevance)**.**

(i) For $0 < p < 1$, define $n_p^\sharp$ by

$$R^{\mathrm{demo}}(n_p^\sharp) = (1 - p^2)R^{\mathrm{demo}}(0).$$

Then $\mathrm{ESS}_0 < n_p^\sharp$ at every finite c, and $\mathrm{ESS}_0 \to n_p^\sharp$ as $c \to \infty$. As $a_0 \to \infty$ with r fixed, $\mathrm{ESS}_0 \to p^2(1 - r)d$ and $n_p^\sharp \to p^2 d$. For $p = 0$, both ESS and the ceiling are zero.

(ii) With d, a_0, and c fixed, as $n \to \infty$,

$$\begin{cases} \mathrm{ESS}_n \to c - a_0^{-1}, & p = 1, \\ \limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c - a_0^{-1}), & p < 1. \end{cases}$$

Specificity cannot remove a relevance error floor. For fixed $p < 1$, Equation (3) gives a positive risk floor $(1 - p^2)a_0 d$, even as $c \to \infty$. Hence zero-demonstration ESS has a finite ceiling. At $p = 0.9$, $d = 16$, and $a_0 = 10$, increasing c from 1 to 100 raises ESS only from 12.0 to 13.5, compared with 15.6 to 117 when $p = 1$ (Figure 2). The finite-a_0 ceiling is approximately 13.5; the diffuse-baseline approximation is $p^2 d = 12.96$.

Demonstrations distinguish low precision from uncertain relevance. Without demonstrations, an unreliable description has the same risk as a reliable description that leaves variance fraction $1 - p^2(1 - r)$. For example, $c = 100$, $p = 0.9$ and a reliable description with $c \approx 0.52$ both have

ESS approximately 13.5 in this setting. Their values diverge once demonstrations are available: after 100 demonstrations, the former has ESS 86.6 and the latter 0.5. Evidence can support the high-precision component of the unreliable description; the reliable low-precision description has no such component to recover.

Treating every description as relevant can be harmful. A fully trusting predictor always uses μ_1, even when the description is irrelevant. Without demonstrations, it is worse than ignoring the description when $p < 1/2$, provided the description is informative ($r < 1$). For fixed n, its risk tends to $2(1-p)a_0 d$ as $c \to \infty$, whereas the demonstrations-only risk tends to zero as n grows. Thus a sufficiently precise, occasionally irrelevant description can outweigh useful demonstrations. At $p = 0.9$ and $c = 100$, full trust is worse than ignoring the description from 15 demonstrations onward in the reported evaluation (Appendix A.4).

5. Meta-Trained Transformers

We first test whether Transformers can learn to use descriptions in the setting where the Bayesian model is correctly specified. Following Garg et al. (2022), we train from scratch on linear regression tasks with $d = 5$ and $a_0 = 10$. A prompt contains an optional description token carrying m, up to 20 demonstration tokens, and a query token (Huang & Ge, 2025). The token does not specify c or p; the model must infer these properties from its training distribution. Half the training prompts omit the description, providing a demonstrations-only learning curve for each network. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, with two or three seeds per configuration. Architecture, optimization, and additional experiments are reported in Appendix B.

Matched training and test distributions. With no demonstrations, the $c = 10$, $p = 1$ networks achieve mean ESS 14.3, compared with a Bayesian ESS of 16.1 on the same prompts (Figure 3a). At lower reliability, and for $c = 1$, the zero-demonstration ESS is within 0.3 of the Bayesian value. The networks also capture the increased value of a partial description as demonstrations accumulate: with $k = 3$ of $d = 5$ directions described at $c = 10$, ESS rises from 3.27 at $n = 0$ to 5.55 at $n = 10$, compared with Bayesian values of 3.22 and 5.64.

Reliability shifts. We also evaluate each network at reliabilities different from its training value. We compare it with a *train-matched* Bayesian predictor, which retains the training reliability, and a *calibrated* predictor, which uses the test reliability (Raventós et al., 2023). At $n = 0$, the Bayesian prediction is $p\, m^\top x$. We therefore estimate a network's *implied trust* as the coefficient q that best approximates its predictions by $q\, m^\top x$, following coefficient-based analyses of learned predictors (Akyürek et al., 2023; Panwar et al., 2024).

For the $c = 10$ description, networks trained at $p = 0.9$ retain implied trust near 0.9 even when every test description is relevant, falling short of the calibrated predictor (Figure 3b). This is consistent with the information available: without demonstrations, m itself does not reveal the test reliability. Networks trained at $p = 1$ instead continue to trust descriptions fully. When 10% of test descriptions are irrelevant, their prediction error is higher with the description than without it from six demonstrations onward, in every seed. Training at $p = 0.99$ avoids this failure in the tested range: descriptions continue to reduce error at every evaluated n. Exposure to occasional irrelevance during training therefore changes how these networks combine descriptions with later evidence.

6. Pretrained Language Models

The meta-trained networks learn description statistics from repeated tasks. We next ask whether pretrained language models use the same information when it is stated explicitly. We use numerical mean estimation, a scalar version of the regression problem with fixed input $x = 1$, following controlled tests of stated priors and in-context evidence (Gupta et al., 2025; Akata et al., 2026), and then a semantic classification task.

6.1. Task and Evaluation

Each task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$ and demonstrations drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, then rounded to integers. A description supplies a value m and uncertainty τ, for example: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." The underlying Gaussian construction has $w \mid m \sim \mathcal{N}(m, \tau^2)$. We express precision in demonstration units as $c = \sigma^2/\tau^2$: stated standard deviations 60, 30, and 10 correspond to $c = 1$, 4, and 36.

We vary the stated precision, and add conditions stating a 50% probability that the description is wrong. In these conditions the description remains relevant; the intervention tests how models respond to the stated probability. We also hold m and the demonstrations fixed while describing a value generated at $c = 36$ as having $c = 4$. This isolates sensitivity to the stated uncertainty from sensitivity to the numerical value itself. Appendix C includes genuinely irrelevant descriptions and evaluations under actual mixtures of relevant and irrelevant descriptions.

The main comparison uses the same 200 tasks for five models with reasoning enabled and disabled: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4

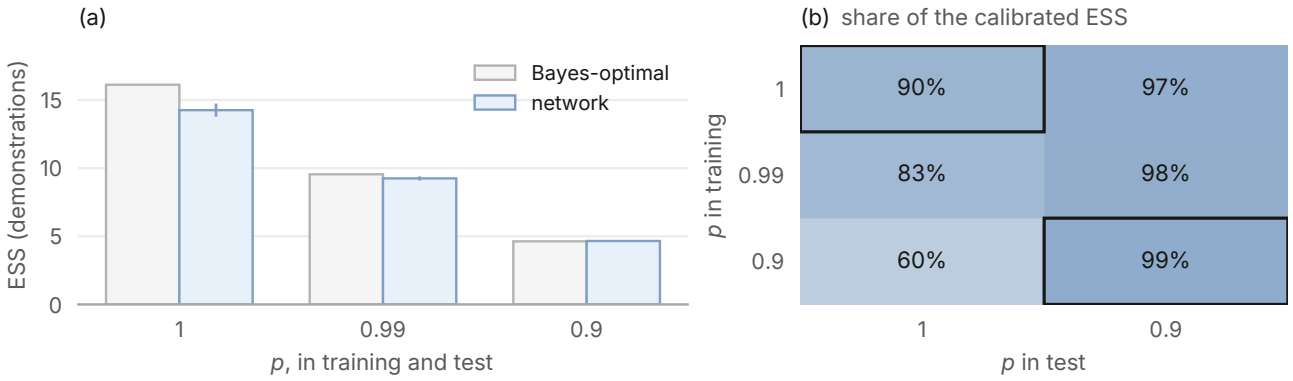


Figure 3. **Meta-trained Transformers approach the Bayesian ESS and inherit their trust from training.** Results for $c = 10$, $d = 5$, $a_0 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own demonstrations-only curve; bars show the mean and one standard deviation across two or three training seeds. The Bayesian comparison uses the same evaluation prompts. (b) Reliability changes at test time. Cells show network ESS as a fraction of the calibrated Bayesian ESS; outlined cells have matching training and test reliability. Table 2 also reports the $c = 1$ condition.

31B (Gemma Team, 2026), GPT-6 Sol, and Claude Opus 4.6. Claude Haiku 5.5 is evaluated with reasoning at medium effort. Checkpoints are held fixed within model, although serving environments and decoding settings differ for some comparisons. Appendix C reports these details, response extraction, token limits, reruns, and a larger non-reasoning evaluation on 2,000 tasks.

A reference based on the prompt. The prompt specifies the observation noise and description, but not the task generator's baseline prior. Our *prompt-based Bayesian reference* therefore uses the stated description and a uniform fallback over the three-digit range. For a description asserted to be relevant, ignoring negligible boundary effects, its prediction is

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{4}$$

where $\bar{y}$ is the demonstration mean. Under the continuous, correctly specified Gaussian model with a flat baseline prior, the description has exact ESS c. There is no random-design dimension term because every observation measures the same scalar mean.

This reference follows the prompt even when a stated precision or reliability differs from the data generator. It is therefore a model of inference from the stated information, rather than an oracle for every evaluation condition. When the description may be wrong, its posterior also depends on the chosen fallback; we report sensitivity to that choice.

For each model, we compute ESS from its own demonstrations-only curve and summarize it by the median over $n \in \{1, 2, 4\}$. The same calculation gives reference ESS 34.6 for $c = 36$, rather than the idealized value 36, because of finite-sample error and interpolation. We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m measures reliance on the description; the reference coefficient is $c/(c + n)$ when the description is asserted to be relevant.

6.2. Reasoning and Stated Precision

Weak sensitivity without reasoning. For the $c = 36$ description, non-reasoning ESS ranges from 2.0 to 10.3 across the five models, compared with 34.6 for the reference (Figure 4a). Models generally underweight this precise description, and most respond weakly when its stated precision changes. At $n = 4$, GPT-6 Sol and Opus 4.6 assign weights 0.47 and 0.40 at stated $c = 36$, and 0.17 and 0.20 at stated $c = 4$, compared with reference weights 0.90 and 0.50. When the same value is described as $c = 4$ instead of $c = 36$, reference ESS falls from 34.6 to 13.7. GPT-6 Sol's falls from 10.3 to 2.4, and no other model's changes by more than 2.8 (Figure 4b).

Reasoning recovers precision weighting in several models. With reasoning enabled, Opus 4.6 and Gemma 4 31B reach approximately the reference ESS in both precision conditions; GPT-6 Sol and Haiku 5.5 at medium effort come close. Among diagnostically separated tasks at $n = 4$, the weight implied by an individual reply is within 0.05 of the reference weight in 89–100% of these models' replies (Appendix C). This supports a change in how the description and demonstrations are combined, beyond an improvement in aggregate error alone.

Qwen3.5 9B and gpt-oss-20b also approach the reference precision weights, but arithmetic errors complicate their ESS. Their demonstrations-only curves can rise at larger n, preventing a crossing within the evaluated grid. For these cases, square markers in Figure 4 report sample equivalence using the reference's demonstrations-only curve, a different quantity from learner-specific ESS. Errors with the

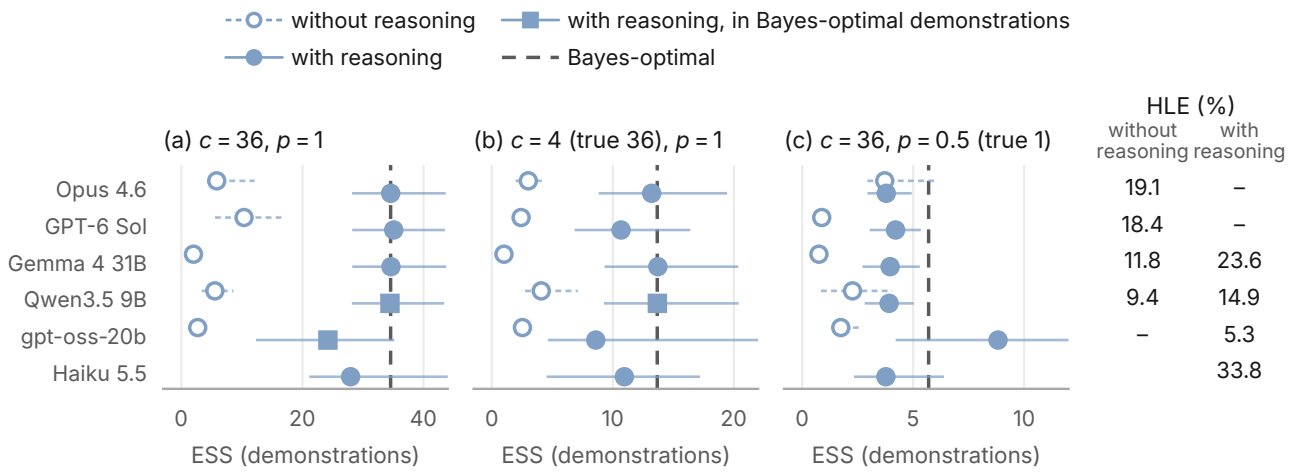


Figure 4. **Reasoning improves the use of stated precision, with model-specific exceptions.** Summary ESS on 200 tasks at $\sigma = 60$: median over $n = 1, 2, 4$, with paired 95% bootstrap intervals over tasks. Open and filled points denote reasoning disabled and enabled; dashed and solid intervals distinguish the modes. Vertical dashed lines show the prompt-based Bayesian reference. (a) Relevant description with $c = 36$: reference ESS 34.6. (b) The same value, generated at $c = 36$, is stated to have $c = 4$: reference ESS 13.7, because the value is more accurate than stated. (c) The description in (a) is assigned a 50% probability of being wrong, while remaining relevant: reference ESS 5.7 under the uniform fallback, which the prompt does not state. Square reasoning markers for Qwen3.5 9B in (a,b) and gpt-oss-20b in (a) use the reference's demonstrations-only curve because the models' own curves do not reach the crossing; these values are in reference-demonstration units. Haiku 5.5 uses medium effort. Intervals are clipped to the displayed range. HLE scores (Phan et al., 2025), as reported by Artificial Analysis (2026), provide descriptive context; dashes indicate unavailable scores for the evaluated mode, and blanks indicate modes not evaluated.

description also reduce gpt-oss-20b's ESS in some conditions. Reported HLE scores do not order these behaviors: some higher-scoring non-reasoning models use precision less accurately than lower-scoring reasoning models. This comparison is descriptive and does not isolate the mechanism responsible for the difference.

6.3. Stated Reliability and Wording

Reasoning models update relevance when the prompt says what a wrong description implies. Adding a stated 50% probability of error lowers reference ESS from 34.6 to 5.7 under the uniform fallback. Without reasoning, no model's ESS falls by more than a factor of 2.7, against about six for the reference, except GPT-6 Sol's, which falls from 10.3 to 0.9. With reasoning, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B closely track a different rule: they retain the stated relevance probability and weight m by $pc/(c+n)$ (Table 9). This averages the relevant and irrelevant predictions using the *prior* mixture weights.

Bayesian inference instead updates relevance by comparing how likely the demonstrations are if the description applies and if it does not. The prompt says only that the mean is then "unknown", which leaves the second likelihood, and so the update, undefined; keeping the stated probability is one defensible response. We therefore repeat the condition for GPT-6 Sol at low effort and Haiku 5.5 at medium effort, changing only the warning to say that the mean "was instead drawn uniformly at random from the whole numbers 100 to 999", the fallback the reference assumes. On tasks where the reference and fixed-rule predictions differ by at least 5, the share of replies closer to the reference rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5 (Appendix C). Once the alternative is specified, both models update relevance from the demonstrations. The update is not always calibrated: with a stated 10% chance, Haiku 5.5 relies on the description less than the reference does, and its ESS falls to 17.9, against 27.5; GPT-6 Sol's is 25.0.

More explicit statistical descriptions are used more consistently. We test three alternative wordings with GPT-6 Sol and Haiku 5.5. Both closely follow Equation (4) under the original wording and "An earlier estimate put its mean at m, with a standard error of τ." Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$," they often return m and disregard demonstrations. With the first informal wording, GPT-6 Sol skips reasoning in 24–51% of replies per condition and usually returns m in those replies. Haiku 5.5 returns m in 71% of the replies for which it does reason.

These alternatives preserve the numerical values but do not specify an equally precise statistical interpretation. The statistical wordings explain how the uncertainty should enter the task; informal uncertainty expressions leave that relationship less explicit. As with the unstated fallback, the

finding therefore concerns the specificity of the description as a whole, including its instructions for interpreting uncertainty, rather than invariance to semantically equivalent phrasing.

6.4. A Semantic Task

Reasoning models use semantic descriptions almost as the reference does. In our semantic task, a company files messages into one of two folders by a hidden policy that requires two of six properties, written in ordinary language, to be present or absent. The prompt states that each of the 60 such policies is equally likely, so the Bayesian reference is exact (Appendix C.7). A specific note states one requirement and names the other property. With reasoning, Haiku 5.5 and GPT-6 Sol value it at 9.9 and 9.5 demonstrations at $n = 2$, against 11.4 for the reference. Given the same note with a stated 50% chance of being wrong and the fallback stated, they update it from the demonstrations. Without reasoning, GPT-6 Sol makes poor use of the note.

7. Related Work

Bayesian accounts of ICL. ICL can be viewed as inference about a latent task (Xie et al., 2022), and Transformers trained on task distributions can approximate posterior prediction (Müller et al., 2022; Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024). Work incorporating additional task information considers input-distribution descriptions (Huang & Ge, 2025), candidate hypotheses (Lin et al., 2025), instructions in generalization bounds (Tong et al., 2026), and prefixes of related datasets for hierarchical Bayesian prediction (Zhu et al., 2026). We focus on a common quantitative question across such settings: how much does task information reduce the number of demonstrations needed for a given prediction error?

Prior effective sample size. ESS is an established way to quantify informative priors (Morita et al., 2008; Neuenschwander et al., 2020). Our measure builds on risk-based definitions (Reimherr et al., 2021) and depends on the observations already available, as does effective current sample size (Wiesenfarth & Calderazzo, 2020). Related work discounts incompatible historical information using predictive loss (Reznik, 2026); robust priors similarly account for possible mismatch (Pichon et al., 2012; Schmidli et al., 2014). Selby et al. (2025) evaluate priors elicited from language models by their predictive sample value. We instead measure descriptions supplied to an in-context learner and compare the value predicted by a task model with the value realized by the learner.

Descriptions and evidence in language models. Controlled studies find that demonstrations can outweigh stated priors (Gupta et al., 2025). Other work examines semantic priors and flipped labels (Wei et al., 2023), weak dependence on demonstration labels (Min et al., 2022), and Bayesian reference curves for in-context prediction (Akata et al., 2026; Arora et al., 2025). We vary stated precision and reliability separately, use fixed-value controls, and compare reasoning modes. Further related work is discussed in Appendix D.

8. Discussion

Description ESS connects two questions: how much task information is available, and how much of it a learner uses. In our model, a reliable description can replace many demonstrations because it constrains the task across directions as well as increasing precision. Unreliability changes this relationship: without demonstrations, even an arbitrarily precise description has bounded value. Demonstrations can then serve a second purpose, providing evidence about whether the description applies. Sufficiently precise partial descriptions can create another form of complementarity, becoming more valuable as examples resolve unspecified aspects of the task.

The empirical results show that these benefits are learnable but not automatic. Meta-trained Transformers approach the Bayesian benchmark under matched task distributions, while their responses to reliability shifts reflect training. Pretrained language models make substantially better use of explicitly stated precision with reasoning enabled. With reasoning, they also update relevance from the demonstrations once the prompt says what a wrong description implies, and they use semantic descriptions of a rule almost as the reference does. When the prompt leaves this open, they keep the stated probability fixed, and when it states uncertainty informally, they often ignore the demonstrations. These findings characterize observable prediction behavior; they do not identify a unique internal mechanism or establish a general separation between understanding and computation.

Scope and limitations. The theory assumes Gaussian linear regression and a particular mixture model of irrelevance. The language-model tasks are small and state their probabilistic information explicitly; they are not unrestricted instruction following. The main comparison uses 200 tasks, and the reasoning-enabled wording test covers two models. Some mode comparisons also differ in serving or decoding settings. The original reliability prompts leave the fallback prior unspecified; we report several choices, and the rerun that states it covers two models. Extending the framework to open-ended semantic constraints, structured tasks, and richer or heavy-tailed description priors (Dalal & Hall, 1983; O'Hagan & Pericchi, 2012) would test how far these relationships extend.

Impact Statement

This work studies how learners combine task descriptions with demonstrations. Its measurement framework may inform the evaluation and design of prompts, especially when descriptions are uncertain or incomplete. The experiments use synthetic prediction tasks. Conclusions about these tasks should not be taken as evidence of reliable instruction following in consequential applications; the broader implications depend on how the framework is extended and used.

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A. Proofs

A.1. Reliable Descriptions

Notation and risk. Write $\varepsilon = 1/a_0$, let $X_n \in \mathbb{R}^{n\times d}$ contain the demonstration inputs as rows, and set $W_n = X_n^\top X_n$. For $\kappa > 0$, define

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. These identities follow from the posterior covariance under Gaussian linear regression. Conditional on the prompt, the posterior mean minimizes squared prediction error, and the independent query satisfies $\mathbb{E}[xx^\top] = I_d$. Averaging its conditional error $x^\top \Sigma x$ therefore gives $\mathbb{E}\,\mathrm{tr}\,\Sigma$.

Throughout the proofs, d is a positive integer, $a_0 > 0$, $c \geq \varepsilon$, $r = 1/(ca_0) \in (0, 1]$, and $p \in [0, 1]$. Noninteger sample sizes refer to linear interpolation between the risks at adjacent integers.

Wishart identities. We use three standard properties of W_n.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \geq d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \frac{d}{n-d-1} \quad (n \geq d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d + 1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$ and the Bayes-optimal ESS is nonnegative.

Proof of Proposition 1. A reliable description without demonstrations has risk

$$R^{\mathrm{desc}}_{p=1}(0) = \frac{d}{c} = ra_0 d.$$

(i) Fixed variance reduction. By (W1), $R^\varepsilon(n) \geq (d-n)a_0$ for $n < d$; the inequality also holds trivially for $n \geq d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{demo}}(t) \geq (d-t)a_0 \qquad (t \geq 0).$$

For $t < (1-r)d$, the right-hand side exceeds $ra_0 d$. The matching sample size must therefore satisfy $\mathrm{ESS}_0 \geq (1-r)d$.

For every integer n,

$$\frac{R^\varepsilon(n)}{a_0} = \mathbb{E}\,\mathrm{tr}(I_d + a_0 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } a_0 \to \infty.$$

Indeed, an eigenvalue $1/(1 + a_0\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real t by interpolation. For fixed $r \in (0, 1)$, the limiting curve crosses rd at the unique point $t = (1-r)d$. Strict monotonicity then gives convergence of the crossing. For $r = 1$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every a_0, giving the same conclusion.

(ii) Fixed precision. For fixed $c > 0$, the description risk d/c is independent of a_0. As $a_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^\varepsilon(n) \longrightarrow \begin{cases} +\infty, & n \leq d+1, \\ \dfrac{d}{n-d-1}, & n \geq d+2. \end{cases}$$

If c is a positive integer, the finite part of this limiting curve equals d/c at the integer $n = c + d + 1$. For every fixed point below this crossing the limiting risk is larger than d/c, and for every fixed point above it the limiting risk is smaller. The crossing of the interpolated finite-a_0 risk therefore converges to $c + d + 1$. The integer condition ensures that the equality occurs at a grid point; linear interpolation does not in general preserve the same expression for noninteger c. □

Large-sample expansion. For fixed d and κ, commuting W_n with κI_d gives

$$(\kappa I_d + W_n)^{-1} = W_n^{-1} - \kappa W_n^{-2} + \kappa^2 W_n^{-2}(\kappa I_d + W_n)^{-1}.$$

The final term is positive semidefinite and bounded above by $\kappa^2 W_n^{-3}$. Using (W2) and (W3),

$$R^{\kappa}(n) = \frac{d}{n - d - 1} - \frac{\kappa d}{n^2} + O(n^{-3}). \tag{5}$$

An additional demonstration reduces the leading term by $d/n^2 + O(n^{-3})$. Thus a fixed change in prior precision and a fixed change in sample size affect risk at the same order. This observation underlies the large-n ESS limits below.

A.2. Unreliable Descriptions

Let $\mathcal{P}_n = (D_n, m, x)$ denote the full prompt, including the query input, and write $\pi_n = P(Z = 1 \mid \mathcal{P}_n)$. Conditional on relevance, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on irrelevance, m is independent of w, and

$$\Sigma_0 = (\varepsilon I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

The relevance probability is obtained from the two Gaussian marginal likelihoods,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + c^{-1} X_n X_n^\top\big)\,, \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + a_0 X_n X_n^\top\big)\,, \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query does not enter this update because it is independent of Z. Without demonstrations, both marginal likelihoods equal one and $\pi_0 = p$: the distribution of m is the same under both hypotheses. At $p = 0$ or $p = 1$, the formula reduces to the corresponding single component.

Proof of Proposition 2. The posterior mean prediction is

$$\mathbb{E}[w^\top x \mid \mathcal{P}_n] = \big(\pi_n \mu_1 + (1 - \pi_n)\mu_0\big)^\top x.$$

Applying the law of total variance over Z yields

$$\begin{aligned} \mathrm{Var}(w^\top x \mid \mathcal{P}_n) = \pi_n x^\top \Sigma_1 x + (1 - \pi_n) x^\top \Sigma_0 x \\ + \pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2. \end{aligned}$$

For any integrable function g of the prompt,

$$\mathbb{E}[\pi_n g(\mathcal{P}_n)] = \mathbb{E}[Z g(\mathcal{P}_n)] = p\,\mathbb{E}[g(\mathcal{P}_n) \mid Z = 1].$$

Conditional on $Z = 1$, the prompt follows the reliable-description model. Conditional on $Z = 0$, the description provides no information about the task. Taking expectations in the variance decomposition therefore gives

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1 - p)R^{\mathrm{demo}}(n) + \mathcal{V}_n,$$

with $\mathcal{V}_n$ as in Equation (2).

At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and the prediction is $p\, m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = a_0, \qquad \mathrm{Var}(m_i) = (1-r)a_0, \qquad \mathrm{Cov}(w_i, m_i) = p(1-r)a_0.$$

It follows that

$$\mathbb{E}(w_i - pm_i)^2 = a_0 - 2p^2(1-r)a_0 + p^2(1-r)a_0 = \big(1 - p^2(1-r)\big)a_0.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction risk is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (3). □

Proof of Corollary 3. (i) No demonstrations. For $0 < p < 1$ and finite c,

$$R^{\mathrm{desc}}(0) = \big(1 - p^2(1-r)\big)R^{\mathrm{demo}}(0) > (1-p^2)R^{\mathrm{demo}}(0).$$

The risk on the left decreases to the right-hand side as $c \to \infty$, equivalently $r \to 0$. The interpolated baseline risk is continuous, strictly decreasing, and tends to zero. Consequently, the matching sample size is strictly below the unique finite $n_p^\sharp$ defined in the corollary and converges to it as $c \to \infty$. At $p = 0$, the description is always irrelevant, so $\mathrm{ESS}_0 = n_0^\sharp = 0$; the non-strict bound remains valid.

The baseline lower bound and normalized limit established in Appendix A.1 also give

$$\mathrm{ESS}_0 \geq p^2(1-r)d, \qquad \mathrm{ESS}_0 \longrightarrow p^2(1-r)d$$

when $a_0 \to \infty$ with r fixed. Applying the same argument to the level $1 - p^2$ gives $n_p^\sharp \to p^2 d$. These are broad-prior limits, not exact identities at finite a_0.

For comparison, an oracle told Z has zero-demonstration risk

$$\big(1 - p(1-r)\big)a_0 d \geq (1-p)a_0 d.$$

Its limiting ceiling is therefore pd demonstrations as $a_0 \to \infty$. The additional reduction from p to p^2 in the non-oracle expression is the cost of uncertainty about relevance.

(ii) Many demonstrations. Fix d, a_0, and c, and define the oracle risk

$$L(n) = pR^c(n) + (1-p)R^\varepsilon(n).$$

Equation (5) implies, uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\left[p(c-\varepsilon) - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. Let $n + \eta_n$ be the sample size at which the baseline risk equals $L(n)$. Then

$$\eta_n = p(c-\varepsilon) + O(n^{-1}).$$

When $p(c-\varepsilon) = 0$, the two risks coincide and $\eta_n = 0$ exactly. If $p = 1$, the description risk equals $L(n)$, so

$$\mathrm{ESS}_n \longrightarrow c - \varepsilon = (1-r)c.$$

If $p < 1$, the variance decomposition gives $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$. Since the baseline risk decreases with sample size, $\mathrm{ESS}_n \leq \eta_n$, proving

$$\limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c-\varepsilon) = p(1-r)c.$$

The nonnegative term $\mathcal{V}_n$ is why the result is an upper bound for unreliable descriptions. □

A.3. Partial Descriptions

A reliable description may constrain only part of a task. Suppose it leaves a fraction r of the baseline variance in $k < d$ directions and leaves the remaining $d-k$ directions unchanged. Its conditional prior precision is c in the first set of directions and $\varepsilon = 1/a_0$ in the second. Write

$$\Lambda = \operatorname{diag}(cI_k, \varepsilon I_{d-k}).$$

The prediction risk is $\mathbb{E}\operatorname{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k < d$ be an integer.*

(i) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1-r)k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.
(ii) For fixed d, a_0, and c, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - 1/a_0) = \frac{k}{d}(1-r)c.$$

In the broad-prior limit, the unconstrained directions dominate the initial error. Even an arbitrarily precise description of $k < d$ directions therefore replaces only about k demonstrations when used alone. After many demonstrations, its value instead approaches the added precision averaged across all d directions. These two limits permit complementarity: a sufficiently precise partial description can be worth more alongside demonstrations than by itself. The numerical example in Table 1 shows this effect. At $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36 with 100 demonstrations.

Proof of Proposition 4. (i) No demonstrations. The description risk is

$$\operatorname{tr}\Lambda^{-1} = (kr + d - k)a_0.$$

Using $R^{\mathrm{demo}}(t) \ge (d-t)a_0$ gives $\mathrm{ESS}_0 \ge (1-r)k$. The normalized baseline risk converges to $(d-t)^+$, locally uniformly in t, by the integer limits and linear interpolation. With r fixed, the normalized description risk is $d-(1-r)k$. With c fixed, it is

$$d - k + \frac{k}{ca_0} \longrightarrow d - k.$$

Because $k < d$, both limiting levels are positive and have unique crossings, at $(1-r)k$ and k, respectively. Convergence of the crossings proves the claim. When the description leaves the prior unchanged, its ESS is identically zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E}W_n^{-2} = \frac{\mathbb{E}\operatorname{tr}W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\operatorname{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\operatorname{tr}(MB)| \le \|M\|_{\mathrm{op}} \operatorname{tr} B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-1} \le d\|\Lambda\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\operatorname{tr}(\Lambda + W_n)^{-1} = \frac{d}{n-d-1} - \frac{\operatorname{tr}\Lambda}{n^2} + O(n^{-3}).$$

Matching this expansion to the baseline risk, as in Corollary 3(ii), gives

$$\mathrm{ESS}_n \longrightarrow \frac{\operatorname{tr}\Lambda - d\varepsilon}{d} = \frac{k(c-\varepsilon)}{d}.$$

□

The same expansion applies to any fixed Gaussian description precision $\Lambda \succeq \varepsilon I_d$ compatible with the baseline task distribution. Its large-sample ESS is the added precision averaged across directions, $(\operatorname{tr}\Lambda - d\varepsilon)/d$.

Table 1. **Demonstrations can increase the value of a precise partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $a_0 = 10$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(1 - r)ck/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.23.

	loose, $c = 1$					specific, $c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

A.4. The Fully Trusting Predictor

Without demonstrations, a predictor that always treats the description as relevant returns $m^\top x$. The moments used in the proof of Proposition 2 give

$$R^{\mathrm{trust}}(0) = \mathbb{E}\|w - m\|^2 = \big(1 - (2p - 1)(1 - r)\big)a_0 d.$$

For $r < 1$, this risk exceeds the baseline $a_0 d$ exactly when $p < 1/2$. At $r = 1$, the description carries no information, $m = 0$ almost surely, and the two risks coincide for every p.

With demonstrations, the fully trusting predictor uses $\mu_1^\top x$ regardless of Z, where

$$\mu_1 = m + (cI_d + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

For fixed a_0 and n, $\mu_1 \to m$ in mean square as $c \to \infty$. Conditional on relevance, $\mathbb{E}\|w - m\|^2 = r a_0 d \to 0$; conditional on irrelevance, it is $(2 - r)a_0 d \to 2a_0 d$. Thus

$$\lim_{c \to \infty} R^{\mathrm{trust}}(n) = 2(1 - p)a_0 d.$$

For every $p < 1$, this is a positive error floor at fixed n, whereas $R^{\mathrm{demo}}(n) \to 0$ as $n \to \infty$. Hence sufficiently many demonstrations can outperform full trust in a sufficiently precise unreliable description. The order of limits matters: this argument takes $c \to \infty$ at fixed n and does not assert that demonstrations can never overcome a fixed finite prior precision.

B. Additional Transformer Results and Training Details

Evaluation. We evaluate each network on 20,000 fresh prompts. At $d = 5$, the demonstration-only learning curve is evaluated at every integer from 0 to 20. We compute ESS at the first crossing of this curve with the description-conditioned error, as in Definition 1. Table 2 reports results at $n = 0$, including evaluation at reliabilities that differ from the training distribution.

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). We train with Adam at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is the posterior mean, which is also optimal under squared error; evaluation uses the predicted mean and the loss in Equation (1).

Each demonstration token contains an input and its label. Two indicator coordinates distinguish token types, and the model uses no positional encoding. When a description is absent, its token is masked out of attention. The prediction is read from the query token. Labels and description values are divided by $\sqrt{a_0}$, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments below, which use 0 to 15.

Without a description, every network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$. Averaged over the demonstration counts used in training, its error is within 0.3 of the Bayesian reference.

Table 2. **Transformer ESS and implied trust under matched and shifted reliability.** Results use $d = 5$, $a_0 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. "Network" reports mean ESS across the indicated training seeds, with standard deviation; for a single seed, the uncertainty is the standard error over prompts. The train-matched Bayesian predictor uses the training reliability, and the calibrated predictor uses the test reliability. We show a single reference value when they coincide. Calibrated values, and coincident train-matched values, use 200,000 prompts, with standard errors at most 0.1. Other train-matched values use the network's test prompts; sampling variation can change these estimates by up to 0.6, so an estimated train-matched value can exceed its calibrated counterpart. "Trust" is implied trust averaged across seeds.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Train-matched	Calibrated	Trust
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

Additional specificity levels. Four additional networks cover $c = 0.2$ and $c = 2$, each at $p = 1$ and $p = 0.9$. Their zero-demonstration ESS values are within 0.03 of the Bayesian values: 2.64 versus 2.66, 2.16 versus 2.15, 7.32 versus 7.35, and 4.38 versus 4.37, respectively.

Higher dimension and partial descriptions. At $d = 10$, we train three seeds with the specific description and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. The network ESS is 17.8, compared with a Bayesian value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$, the network captures most of the description's value but falls short of the reference.

We also train networks with a partial description, as defined in Appendix A.3. The description token contains m in the first $k = 3$ of the $d = 5$ coordinates and zeros elsewhere; the remaining task coordinates follow the baseline prior. For this $c = 10$ description, mean ESS across three seeds increases from 3.27 at $n = 0$ to 5.55 at $n = 10$. The corresponding Bayesian values are 3.22 and 5.64. This increase agrees with the behavior described by Proposition 4.

ESS with demonstrations. Figure 5 evaluates each network at its training reliability. For the less precise description, network ESS remains within 0.6 of the Bayesian value at every demonstration count whose crossing lies within the evaluation grid. For the more precise description, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. At $n = 5$, the three networks trained at $p = 0.9$ have ESS values of 4.83, 4.79, and 4.77, compared with the Bayesian value of 5.20.

Reliability shift after demonstrations are observed. For the specific description, networks trained at $p = 1$ and $p = 0.99$ both behave approximately as if the description were certainly relevant when $n = 0$. Their behavior separates once demonstrations are available. At test reliability $p = 0.9$, the description reduces error for the networks trained at $p = 0.99$ at every evaluated n, as it does for the Bayes-optimal predictor. For networks trained at $p = 1$, the description instead increases error from six demonstrations onward, as reported in Section 5.

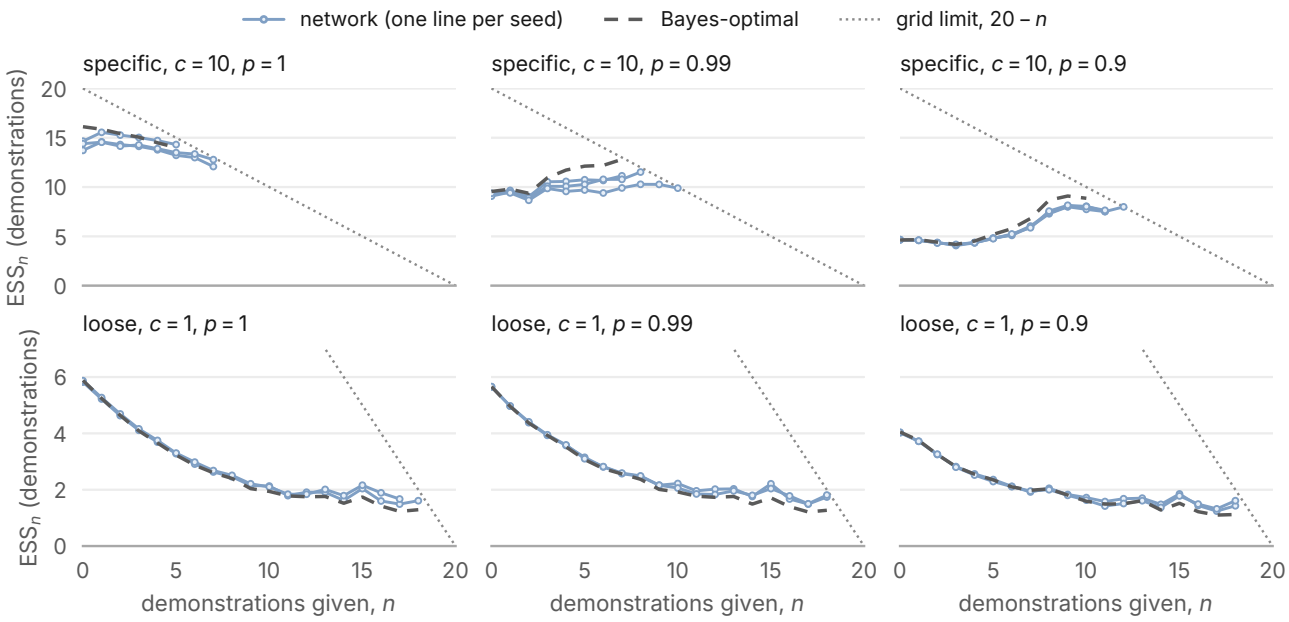


Figure 5. **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayesian reference is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

C. Additional Language-Model Results and Evaluation Details

We generate 2,000 tasks. Six open-weight models answer all tasks without reasoning; we refer to these experiments as the *full run*. The models are gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B, which also appear in Section 6, together with Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. The experiments in Section 6 use the first 200 tasks. Unless stated otherwise, the results in this appendix use the full run without reasoning.

Table 3 reports summary ESS, defined as the median over $n \in \{1, 2, 4\}$, with uncertainty intervals. Figure 6 shows the underlying learning curves. Tables 11 and 12 report ESS at every evaluated demonstration count; Table 4 compares interpolation methods; Table 13 reports description weights for the models with a distribution readout; and Table 5 evaluates genuinely unreliable descriptions.

Throughout this appendix, the *prompt-based Bayesian reference* uses the probabilistic claims in the prompt together with an explicit assumption for the unspecified fallback prior. It is optimal under that assumed model. When the stated description differs from the data-generating process, it need not minimize error under the evaluation distribution. The reference labeled "Bayes-optimal" in the plots and tables has this meaning.

C.1. Prompts, Tasks, and Model Readouts

Model inclusion. We retain a model only if it produces a number in at least three quarters of replies overall and in at least half of replies without a description at each of $n = 1$, 2, and 4. The latter conditions determine the baseline curve used for summary ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction.

The full run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are sampled by a seeded generator. Reasoning is disabled where the chat template supports this option. For gpt-oss-20b, generation begins in the final channel, after the channel normally used for reasoning.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution over the 900 three-digit number tokens. We call this the *distribution readout*. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Tables 3 and 6). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Table 3. **Description ESS and weights in the full run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or accompanied by a stated 10% or 50% chance of being wrong. Brackets give 95% bootstrap intervals. An upper bound, $\leq$, indicates that one demonstration alone outperforms the description-conditioned prediction. Right: the coefficient on the stated value at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Bayes-optimal	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.

548
431

Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." In the rerun of Section 6 that states the fallback, its last clause becomes "in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description value is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or description value falls outside 100 to 999; this occurs for less than 1% of tasks. The full run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table 11).

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 6, 11, 12, and 13 report this quantity as "number". Tables 6 and 13 italicize cells whose rate is below three quarters.

Written-readout implementation and checks. The written readout extracts the first number with three digits before an optional decimal part and retains the decimals. Without reasoning, decoding is greedy with a limit of eight new tokens. Those predictions were recorded using an earlier readout that retained only the integer part; this rarely changes the value because decimal outputs are uncommon. With reasoning, extraction is applied to the final answer. Replies without a qualifying number are excluded from both error curves and weight regressions.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout summary ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and

2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 63% of written replies across all conditions, so Table 3 reports only its distribution readout. In the wording experiment, however, its numeric response rate exceeds three quarters for the original wording. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

C.2. ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 1 to squared error against the hidden mean. In the full run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. We interpolate squared error, rather than its square root, linearly in $\log n$. The ESS at n is the first interpolated demonstration count t that matches the description-conditioned error, minus n.

If the description-conditioned error lies below the entire baseline curve up to $t_{\max}$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we report $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. Summary ESS uses $n = 1$, 2, and 4. We exclude $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error ranges from 9 to 23, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the description-conditioned and baseline curves and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below.

Reference prior and exact sample equivalence. The prompt-based Bayesian reference takes "unknown" to mean uniform over 100 to 999. Without a description, its prediction is therefore approximately the demonstration mean, with negligible boundary truncation for the evaluated prompts. For a relevant Gaussian description, the idealized continuous model with a flat baseline prior has exact sample equivalence c. Integer rounding, finite-range truncation, and the task-rejection rule are not part of that exact identity; we compute the reported reference values on the same generated tasks as the models. When a chance of being wrong is stated, the reference uses it as a prior probability of irrelevance and updates that probability from the demonstrations. Otherwise, it treats the description as relevant.

Applying the empirical ESS procedure to the full-run reference gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the 200 tasks of Section 6, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $(1 - r)c$ in Section 3. We use the prompt-based reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Sections 3–4, a reference that accepts an incorrect description can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The full-run grid doubles n, so interpolation can affect ESS. On the reference's idealized errors, σ^2/n without a description and $\sigma^2/(c + n)$ with one, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, the relevant crossings are unique. Among the twelve baseline curves across the two noise levels, only Phi-4 at $\sigma = 60$ increases, slightly, from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 4 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using pool adjacent violators; and a relative-least-squares fit of $A/(n + n_0) + B$ to the baseline errors. The final form is a Bayesian learning curve plus an error floor and recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's summary ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and

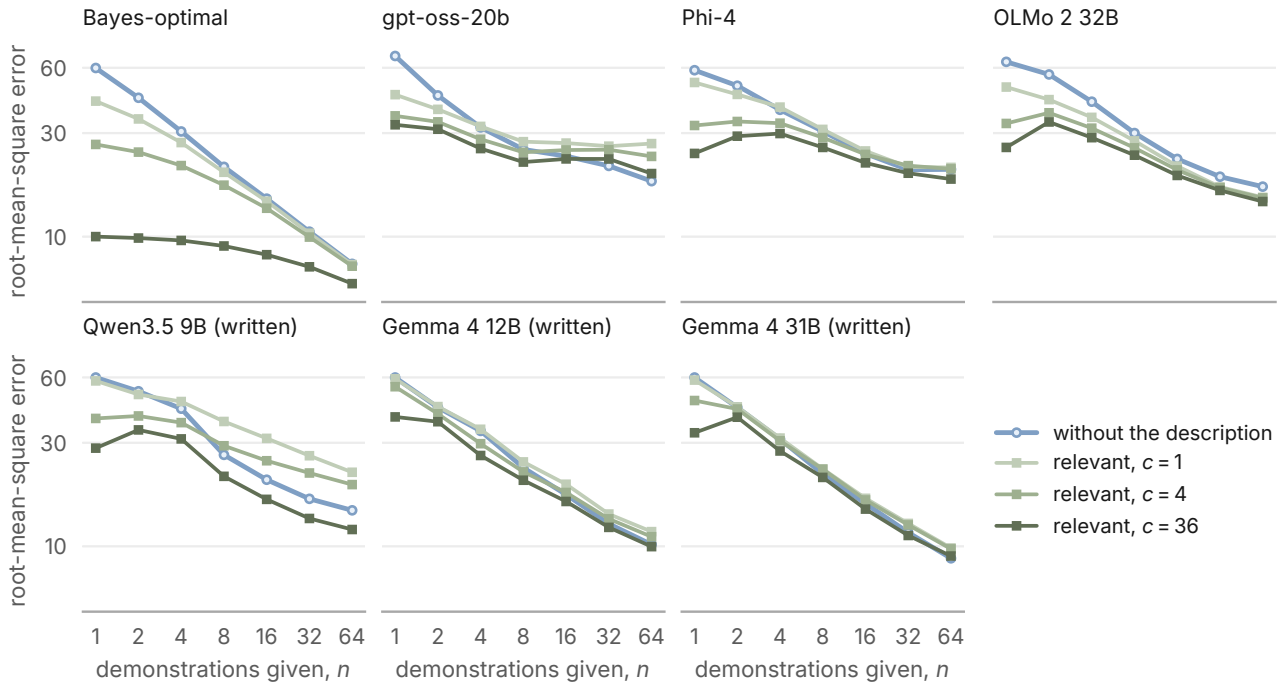


Figure 6. **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the full run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the description-conditioned point at n to the baseline curve at the same error. For four models, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does (Table 13).

$c = 36$ has ESS 17 under the estimator in Table 11, but 9.2 under the fitted curve.

C.3. Unreliability, Description Accuracy, and Wording

Irrelevant descriptions. An irrelevant description uses the value from a different, independently sampled task. At $c = 36$, these descriptions have negative ESS for every model and for the prompt-based reference, which treats them as relevant. The models assign less weight to the description and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference.

Actual unreliability. The stated-unreliability experiments keep descriptions relevant and vary what models are told. Table 5 instead evaluates the mixture in Section 2. We combine squared errors from the relevant and irrelevant conditions, on the same tasks, with weights p and $1 - p$, and then estimate ESS. We evaluate models both with and without disclosure of the irrelevance probability. The fully trusting reference always treats the description as relevant; the calibrated reference starts from irrelevance probability $1 - p$ and updates it from the observations.

At actual reliability $p = 0.9$, a $c = 36$ description has ESS 5.5 for the calibrated reference and 0.2 for the fully trusting reference. Model ESS ranges from 0.4 to 3.2 when the reliability is not stated and from -0.1 to 4.0 when it is stated. Model weights depend weakly on p, and stating the probability changes ESS by at most two demonstrations in either direction. At $p = 0.5$, the description is worth less than one demonstration to the calibrated reference and at most one to every model.

Wording without reasoning. We repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 6 uses the full-run tasks and written readouts.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. A coefficient above 1 means that predictions vary more strongly with the stated value than the value itself. Nevertheless, none

Table 4. **Sensitivity of summary ESS to baseline-curve estimation.** Median ESS over $n = 1, 2$, and 4 for relevant descriptions, using the full run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Bayes-optimal	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

Table 5. **Description ESS under actual unreliability.** Relevant and irrelevant $c = 36$ conditions are combined with weights p and $1 - p$. Results are summary ESS with 95% task-bootstrap intervals for the full run without reasoning at $\sigma = 60$. The fully trusting reference assumes relevance; the calibrated reference uses the actual reliability and updates it from demonstrations. A bound indicates that one demonstration alone outperforms the description-conditioned prediction at one or more summary demonstration counts.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Fully trusting	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Calibrated	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	−0.1 [−0.1, 0.0]	−0.1 [−0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	−1.0 [−1.0, −0.7]	−0.1 [−0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

of the tested wordings yields the reference's sensitivity to specificity: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison also varies how precisely the task is specified.

Separating stated specificity from description accuracy. For each task, we take a description value generated either at $c = 36$ or at $c = 1$ and present it with each stated specificity in $\{1, 4, 36\}$. The tasks and demonstrations match the full run, and the experiment includes its own no-description prompts at every n (Table 7). Repeated conditions reproduce the full run exactly except for gpt-oss-20b. Its distribution readout changes by a median of 0.8 to 1.2 across conditions, with a maximum difference of 39; its written replies change in 12% of cases, with median difference 0. A newer library version is a possible explanation.

The prompt-based reference accepts the stated specificity. A value generated at $c = 1$ but stated as $c = 36$ therefore has reference ESS of about -0.8; a model that discounts the stated specificity can outperform it in that misspecified condition. For models without reasoning, description accuracy primarily determines ESS. When the value is generated at $c = 1$, each model's ESS remains between -0.1 and 2.0 across all stated specificities.

Table 6. **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the full-run tasks. "Number" denotes the numeric response rate; italics mark rates below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated specificity remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

C.4. Reasoning Runs and Execution Settings

Shared evaluation prompts. Every model in Figure 4, as well as Olmo 3 7B Think (Team Olmo, 2025), is evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1, 2, 4, \ldots, 64$, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b run omits the two additional points and contains 7,400 prompts. Results without reasoning for the open-weight models are taken from the full run restricted to these tasks. The two modes receive the same prompts, although their serving environments can differ as detailed below. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 loop to the token limit. Each is rerun once using the model card's recommended reasoning sampling settings; 10 remain unfinished.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun using the model card's recommended sampling settings.

Olmo 3 7B Think runs in vLLM with greedy decoding and at most 32,768 new tokens. We rerun the 169 replies that reach this limit once with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model uses 9,215 reasoning tokens on average. Its long greedy outputs are sensitive to hardware: an earlier stated-specificity run on a different GPU produces the same final number on only half the prompts. We therefore report one run. Despite its extensive reasoning, this model learns little from the demonstrations; its prediction behavior is reported below.

Table 7. **Sensitivity to stated specificity with the value and demonstrations held fixed.** Results use $\sigma = 60$, the full-run tasks, and the readout conventions of Table 3. For a value generated at $c = 36$, the table reports its weight at $n = 1$ and summary ESS under each stated c. Results for the 200-task runs of Section 6 appear in Table 8.

	weight, $n = 1$			ESS		
stated c	1	4	36	1	4	36
Bayes-optimal	0.50	0.80	0.97	2.5	13	36
gpt-oss-20b	0.48	0.47	0.49	2.7	2.6	2.8
Phi-4	0.81	0.75	0.74	12	7.2	7.6
OLMo 2 32B	0.70	0.70	0.70	4.8	4.6	5.3
gpt-oss-20b (written)	0.52	0.46	0.50	2.9	3.5	3.8
Qwen3.5 9B (written)	0.76	0.75	0.80	3.3	3.5	4.0
Gemma 4 12B (written)	0.38	0.29	0.59	1.0	1.0	1.8
Gemma 4 31B (written)	0.43	0.49	0.72	0.2	0.3	1.4

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`) runs through the Codex command-line agent under a subscription, in a read-only sandbox, with an empty configuration directory, so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate reasoning disabled, with no reasoning tokens, and reasoning enabled at low effort. An earlier configuration loaded user instructions and allowed shell commands, which the model used in about 8% of replies, mostly to average long demonstration sequences. Without reasoning, that configuration also gave the description much less weight (ESS 1.5 rather than 10.3 at $c = 36$); with reasoning, every summary ESS changes by at most 2.7. The wording runs below used the earlier configuration, and the model invoked a shell in 4 of their $5{,}400$ replies.

Claude Opus 4.6 and Haiku 5.5 run through the Claude Code command line under a subscription, with one prompt per call, no tools, no project or user instructions, and a one-line system prompt. Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled; the disabled mode emits no reasoning tokens. When enabled, it reasons before at least 99% of replies in each description condition and averages 599 reasoning tokens across all replies. Without a description, it reasons before only 1%, 24%, and 79% of replies at $n = 1$, 2, and 4, respectively. Haiku 5.5 is evaluated at medium and low effort, using the command line's default sampling temperature.

Interpreting the comparison between modes. The experiments hold the model checkpoint and prompts fixed within each paired comparison, but reasoning settings, serving software, readout, retry policy, and access to calculation tools are not uniform across all runs. The results describe the behavior of the evaluated configurations. They do not by themselves isolate a single internal mechanism responsible for the difference between modes.

ESS when baseline arithmetic fails. With reasoning, gpt-oss-20b and Qwen3.5 9B assign approximately the reference weight to the description: at $n = 4$ and $c = 36$, their weights are 0.89 and 0.90, compared with 0.90 for the reference. However, both make errors when averaging longer demonstration sequences. Qwen3.5 9B has no-description root-mean-square error 15.1 at $n = 16$ and 35.4 at $n = 32$, compared with reference errors 14.8 and 10.0. From $n = 32$ onward, between 4 and 24 replies per cell differ from the sample mean by more than 20; one reply gives 945 for numbers averaging 467.

For these models, the baseline curve rises after 16 demonstrations, and some description-conditioned errors never meet it within the evaluated grid. Table 8 therefore also reports, in parentheses, a comparison using the Bayesian reference's baseline curve. This quantity is the number of demonstrations required by the reference without a description to match the model's error, rather than ESS in the model's own units. The gpt-oss-20b model also makes arithmetic errors with descriptions, reducing the reported ESS in those conditions; for example, it obtains 1.3 when the prompt states a 10% chance of being wrong.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks, and using either baseline changes their ESS by at most 1. At low effort, Haiku 5.5 often skips reasoning and repeats the demonstration. Its description weight at $n = 1$ is consequently low even in conditions whose ESS is close to the reference value.

Table 8. **Description ESS across reasoning configurations.** Results use the first 200 tasks at $\sigma = 60$. Summary ESS is computed using each run's baseline curve; parentheses show the value using the Bayesian reference's baseline where it differs. "10% wrong" and "50% wrong" denote the stated probability for a $c = 36$ description. The last column uses a value generated at $c = 36$ but described as $c = 4$. "Reasoning" identifies the mode and provider effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 denotes a crossing beyond the grid. Reference values use the same tasks and estimator. Olmo 3 7B Think's no-description error never drops below its error at one demonstration, so its reported crossings require particular caution; using the reference baseline gives a different comparison rather than resolving the weak model baseline.

Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10% wrong	50% wrong	Stated $c = 4$ True $c = 36$
Bayes-optimal	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.7	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	> 60 (21)	3.9	> 60 (14)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	11 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.4 (3.3)	26 (24)	≤ -1.0	27 (26)	5.7 (5.6)	16 (8.1)
Olmo 3 7B Think, with reasoning	≤ -0.7 (≤ -0.9)	-0.7 (-0.9)	> 60 (≤ -0.5)	≤ -1.0	> 60 (0.8)	> 60 (0.8)	> 60 (-0.7)

C.5. Prediction Rules and Reliability Updating

Rules inferred from individual replies. At $n = 4$, we estimate the description weight implied by an individual prediction as $(\hat{y} - \bar{y})/(m - \bar{y})$. We use the 61 tasks per cell where $|m - \bar{y}| \geq 30$ and exclude replies without a number. We classify a reply as consistent with a rule if its implied weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with its two precisions exchanged (0.10 and 0.80); the description counted as one additional observation, $1/(n + 1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 match the reference rule on 89 to 100% of eligible replies. Without reasoning, 79% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think instead repeats the description value in 96 to 97% of replies at $n = 1$, regardless of stated specificity. At $n = 4$, 70 to 77% of its replies match none of the rules, and only 3 to 8% match the reference. Its no-description root-mean-square error remains near 55 from $n = 1$ to 32, compared with reference error 10 at $n = 32$. Reasoning mode is therefore not sufficient for the desired behavior in every tested model.

A fixed reliability rule. Table 9 reports description weights for $c = 36$ when the prompt states a probability of being wrong. A predictor that combines the description-conditioned and no-description predictions with fixed weights p and $1 - p$ assigns weight $p\,36/(36 + n)$ to m. Under the original warning, which leaves the fallback unstated, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B with reasoning are close to this fixed rule. Opus 4.6 and Haiku 5.5 at medium effort fall below the reference weight by $n = 4$. Haiku 5.5 at low effort, which often skips reasoning at $n = 1$, comes closest to the reference weight. These are behavioral comparisons; they do not identify the internal procedure used in every reply.

Sensitivity to the unspecified fallback prior. The prompt states only that, if the description is wrong, "the mean is unknown". A Bayesian relevance update compares the likelihood of the observations under the description with their likelihood under this fallback. Because the prompt leaves the fallback distribution unspecified, it does not determine a unique Bayesian relevance update.

For the $c = 36$ description with a stated 50% chance of being wrong, the reference's fitted weights on the first 200 tasks at $n = 1$, 2, 4, 8, and 16 are 0.58, 0.61, 0.64, 0.58, and 0.54 with the uniform fallback over 100 to 999. They are 0.49, 0.49, 0.53, 0.49, and 0.45 with the actual task prior $\mathcal{N}(500, 100^2)$; 0.52, 0.54, 0.58, 0.53, and 0.49 with $\mathcal{N}(500, 200^2)$; and 0.72, 0.74, 0.74, 0.67, and 0.60 with a uniform fallback over 0 to 2,000. The fixed rule gives 0.49, 0.47, 0.45, 0.41, and 0.35.

By four demonstrations, the reference weight exceeds the fixed-rule weight for every fallback examined. Thus the observed agreement of several models with the fixed rule persists across these choices of reference. The size of their departure from Bayesian updating depends on the fallback, however; under the actual task prior, the reference update is small before four

demonstrations.

Table 9. **Description weights under stated unreliability.** Coefficients on the $c = 36$ description value for the first 200 tasks, with stated irrelevance probability $1 - p$. "Fixed" uses weight $p\,36/(36 + n)$; the reference updates relevance using the uniform fallback over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition, and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. Several reasoning configurations closely track the fixed rule.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.50	0.46	0.43	0.88	0.84	0.74
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

Stating the fallback. To remove this ambiguity, we rerun the stated-reliability condition on the same 200 tasks, for $c = 36$, stated chances of 10% and 50%, and $n = 1$, 2, 4, with the fallback stated as the reference's uniform over 100 to 999. We use GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with 1,200 prompts each. Table 10 compares their description weights with those of the original runs. With a stated 50% chance, the reference and the fixed rule differ by at least 5 on 123 to 155 tasks per cell. On those tasks, the share of replies closer to the reference than to the fixed rule rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5. With a stated 10% chance, at most 7 tasks per cell separate the two rules, so this share is uninformative there.

Summary ESS, computed against each model's curve without the description from the main run, changes little at 50%: from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5, against 5.7 for the reference, because the two rules give similar errors at $n \leq 4$. At 10%, it falls from 29.7 to 25.0 and from 32.6 to 17.9, against 27.5: with the fallback stated, Haiku 5.5 discounts the description more than the reference does, and GPT-6 Sol slightly more.

Table 10. **Stating the fallback changes the reliability rule.** Coefficients on the $c = 36$ description value for the first 200 tasks, as in Table 9. "Unstated": the original warning, "the mean is unknown"; "stated": the mean is otherwise uniform over 100 to 999, as the reference assumes.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

C.6. Wording with Reasoning Enabled

GPT-6 Sol. We evaluate the three alternative wordings at low effort for the three relevant descriptions and $n = 1$, 2, and 4, using the first 200 tasks. Of 5,400 replies, 5,399 contain a number. Under "An earlier estimate put its mean at m, with a standard error of τ", the model reasons before every reply, and its description weight is within 0.01 of the reference in every cell, as under the original wording.

Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", it skips reasoning in 24 to 51% and 11 to 50% of replies per cell, respectively. Of those replies, 83% and 87% repeat the description value. At $c = 1$ and $n = 1$, the fitted description weights are consequently 0.89 and 1.00, compared with reference weight 0.50. Among replies that do include reasoning,

68% and 43% are within 1 of the reference prediction, while 36% and 68% repeat the description value. These categories overlap at $c = 36$, where the description value and reference prediction can be close.

Haiku 5.5. We evaluate the same alternatives at medium effort; all 5,400 replies contain a number. Under "An earlier estimate put its mean at m, with a standard error of τ", the model reasons before at least 72% of replies in each cell, and 99% of the replies with reasoning lie within 1 of the reference prediction. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", it skips reasoning in 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning, 71% and 91% repeat the description value, while 31% and 20% lie within 1 of the reference prediction. At $c = 1$ and $n = 4$, its description weights are 0.72 and 0.93, compared with reference weight 0.20.

These wording changes affect the precision of the probabilistic specification as well as the model's behavior. All provide a value m and a scale τ, but "give or take" and "$\pm$" do not uniquely identify τ as a prior standard deviation or establish a sampling model. The experiment therefore shows that accurate combination of descriptions and demonstrations under explicit statistical statements does not automatically transfer to these less explicit formulations. It does not isolate sensitivity to semantically equivalent descriptions.

C.7. A Semantic Task

Task. A message has six binary properties: the sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property one way or the other, in one of three phrasings per value, such as "I'm a contractor from another company." or "I'm on the sales team here." A policy uses exactly two properties and requires each to be present or absent; a message goes into folder A when both requirements hold, and into B otherwise. The prompt describes this family of policies and states that the company's policy was chosen at random, with each of the 60 possible policies equally likely. It then gives the policy note, n previously filed messages with their folders, and four new messages, and asks for the probability that each new message goes into A, as a number between 0 and 1. Without reasoning, GPT-6 Sol reads each property correctly in 97 to 100% of 96 messages, so the phrasings are unambiguous.

Policy notes. The broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," which leaves 20 policies. The specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," which leaves 2. The unreliable note adds "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." It is wrong in half the tasks, when it describes an independent policy.

Reference and ESS. The reference's probability of A is the share of policies consistent with the note and the filed messages that send the new message to A. For the unreliable note, it mixes these policies with those consistent with the filed messages alone, weighting the note by the stated 50% updated on the demonstrations. On 3,000 simulated tasks, it is calibrated in every condition, and its posterior probability that the unreliable note is right matches how often it is. The loss is the squared error of the probability against the true folder, averaged over the four messages. ESS follows Definition 1, against each learner's own curve without a note at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32, interpolated linearly; on simulated tasks, this grid moves the reference's ESS by at most 0.6 from its value read at every n.

Runs. We use 200 tasks, each a policy draw, and 19 cells: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4; this gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort and GPT-6 Sol without reasoning and at low effort, through the command-line agents of Appendix C.4, with no tools. No reply called a tool. Of the replies, 14 from GPT-6 Sol at low effort and one from Haiku 5.5 give no probability and are left out. Haiku 5.5 reasons before every reply, with a median of 3,008 reasoning tokens; GPT-6 Sol at low effort reasons before 91%, with a median of 142.

Results. Table 14 reports the ESS of each note. With reasoning, both models are within 2 of the reference in every cell but one, GPT-6 Sol's specific note at $n = 4$: 26.1 against 14.8. Its loss with that note, 0.008, is close to the reference's 0.007, but its own curve without a note flattens past $n = 16$, with loss 0.022 at $n = 20$ against 0.004, so it needs more of its own demonstrations to match the note. For the unreliable note, we compare predictions with the reference and with a fixed rule that keeps the stated 50%, on messages where the two differ by at least 0.05. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5, 77% and 87% for GPT-6 Sol at low effort, and 45% and 45% for GPT-6

Table 11. **ESS by description condition and demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. An upper bound, $\leq$, marks error above the first baseline point; $> 64 - n$ means that the description-conditioned error lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Bayes-optimal	$\sigma = 60, c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60, c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60, c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60, c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60, c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30, c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30, c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30, c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60, c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60, c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60, c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60, c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60, c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60, c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60, c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60, c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60, c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30, c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30, c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30, c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60, c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60, c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60, c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60, c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60, c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30, c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30, c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30, c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60, c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60, c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60, c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60, c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30, c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30, c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30, c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table 12. **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the full run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table 11.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Qwen3.5 9B (written)	$\sigma = 60$, $c = 1$, relevant	1.00	0.2	0.3	-0.9	-2.8	-9.3	-24	-51
	$\sigma = 60$, $c = 1$, irrelevant	1.00	≤ 0.0	≤ -1.0	-2.7	-4.8	-9.7	-23	-49
	$\sigma = 60$, $c = 4$, relevant	1.00	3.9	2.7	1.3	-0.7	-6.2	-18	-44
	$\sigma = 60$, $c = 4$, irrelevant	1.00	≤ 0.0	≤ -1.0	-1.7	-3.8	-9.4	-24	-49
	$\sigma = 60$, $c = 36$, relevant	1.00	6.5	4.0	2.8	6.9	17	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	-2.8	-5.1	-10	-24	-51
	$\sigma = 30$, $c = 1$, relevant	1.00	≤ 0.0	0.2	-0.7	-2.5	-8.7	-22	-49
	$\sigma = 30$, $c = 4$, relevant	1.00	3.4	2.7	1.5	-0.3	-4.5	-14	-40
	$\sigma = 30$, $c = 36$, relevant	1.00	4.7	4.3	3.0	9.2	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	1.00	3.8	-0.4	-0.7	-1.8	-8.5	-21	-47
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-10	-24	-50
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	-0.3	-3.3	-4.2	-6.9
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-5.2	-10	-25	-53
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	1.00	4.5	-0.1	-0.4	-1.9	-9.0	-23	-50
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-26	-54
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	-0.1	-4.0	-4.2	-12
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-7.0	-13	-27	-56
Gemma 4 12B (written)	$\sigma = 60$, $c = 1$, relevant	1.00	0.0	0.0	-0.1	-0.6	-3.2	-5.4	-21
	$\sigma = 60$, $c = 1$, irrelevant	1.00	≤ 0.0	-0.4	-1.3	-1.3	-4.4	-6.8	-24
	$\sigma = 60$, $c = 4$, relevant	1.00	0.3	0.5	1.4	1.1	-0.6	-2.7	-13
	$\sigma = 60$, $c = 4$, irrelevant	1.00	≤ 0.0	-0.4	-1.1	-1.2	-4.1	-6.9	-23
	$\sigma = 60$, $c = 36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	> 0
	$\sigma = 60$, $c = 36$, irrelevant	1.00	≤ 0.0	-0.8	-1.3	-1.3	-4.9	-7.9	-26
	$\sigma = 30$, $c = 1$, relevant	1.00	≤ 0.0	-0.2	-0.1	-0.5	-3.2	-7.9	-27
	$\sigma = 30$, $c = 4$, relevant	1.00	0.7	0.8	2.1	1.5	-0.2	-4.6	-18
	$\sigma = 30$, $c = 36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	-1.8	-8.9
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	-1.0	-3.1	-7.3
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	-0.2	-0.4	-0.5	-2.2	-5.4	-13
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	-0.6	-1.3
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	-0.2	-0.5	-0.5	-2.5	-5.8	-15
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	-0.1	-1.7	-4.5
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	-0.3	-0.4	-0.4	-1.3	-4.1	-9.6
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	-0.3	-0.6	-0.4	-1.9	-4.8	-13
Gemma 4 31B (written)	$\sigma = 60$, $c = 1$, relevant	1.00	0.1	0.0	-0.1	-0.5	-1.4	-5.1	-13
	$\sigma = 60$, $c = 1$, irrelevant	1.00	≤ 0.0	-0.1	-0.1	-0.5	-1.0	-4.5	-11
	$\sigma = 60$, $c = 4$, relevant	1.00	0.8	0.1	0.2	-0.5	-1.0	-4.3	-12
	$\sigma = 60$, $c = 4$, irrelevant	1.00	≤ 0.0	-0.2	-0.2	-0.4	-1.0	-4.4	-10
	$\sigma = 60$, $c = 36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	-2.9
	$\sigma = 60$, $c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	-0.8	-0.6	-1.1	-2.7	-4.8
	$\sigma = 30$, $c = 1$, relevant	1.00	≤ 0.0	0.0	-0.1	-0.7	-1.2	-3.0	-8.1
	$\sigma = 30$, $c = 4$, relevant	1.00	0.5	0.0	0.1	-0.5	-1.3	-3.4	-8.3
	$\sigma = 30$, $c = 36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	-0.4
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	-0.7	-1.9	-5.0	-15
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	-0.1	-0.4	-1.2	-4.1	-9.5
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	-0.3	-1.4	-4.8	-12
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	-0.2	-0.4	-0.9	-3.0	-6.2
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	-0.7	-1.7	-4.6	-17
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	-0.2	-0.4	-1.2	-3.6	-12
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	-0.3	-1.5	-4.5	-15
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	-0.1	-0.4	-1.0	-2.9	-8.5

Table 13. **Description weights by demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is defined as in Table 11; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01

Sol without reasoning. Without reasoning, GPT-6 Sol learns slowly from demonstrations: its loss without a note is still 0.078 at $n = 32$, against 0.000 for the reference. In its own unit, a note can therefore be worth as many demonstrations as to the reference or more, such as 13.5 for the specific note at $n = 2$, although its loss with that note is 0.130, against 0.027.

Table 14. **ESS of each policy note in the semantic task.** ESS on 200 tasks, against each learner's own curve without a note. "≤ 0": the note did not lower the loss; "> 28": the curve without a note does not reach the loss with the note by $n = 32$. Paired 95% bootstrap intervals over tasks for the specific note at $n = 2$: reference $[9.7, 12.9]$, Haiku 5.5 $[8.7, 11.5]$, GPT-6 Sol at low effort $[7.8, 12.7]$, and without reasoning $[3.7, 21.9]$.

Note	n	Reference	Haiku 5.5, medium	GPT-6 Sol, low	GPT-6 Sol, none
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

D. Further Related Work

Descriptions as substitutes for examples. Beyond in-context learning, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) compare optimized prompts with demonstrations in terms of their length and effect. These results motivate measuring descriptions relative to examples, while differing from our predictive-risk comparison within a fixed task distribution and learner.

Text and probabilistic prediction. Text supplied alongside numbers can change numerical predictions (Requeima et al., 2024). Related work studies how closely language-model belief updates follow Bayesian inference and how training can improve probabilistic reasoning (Qiu et al., 2026). He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions. Our experiments use a controlled numerical task with explicit uncertainty so that description precision, relevance, and prediction weights can be compared with a specified probabilistic reference.